

MUHAMMAD HAIDER ZAIDI

☎ +49 1523 1698988 ✉ mhaiderzaidi21@gmail.com in [linkedin.com/in/muhammad-haider-zaidi](https://www.linkedin.com/in/muhammad-haider-zaidi)
🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, SQL, C++, PySpark

Tools/Frameworks: TensorFlow, PyTorch, FastAPI, Docker, Kubernetes, Snowflake, Dask, TensorRT, Azure, AWS SageMaker, CI/CD, Git, Airflow

Competence: Time Series Forecasting, Machine Learning, MLOps, Data Engineering, Feature Engineering, Model Deployment, Experiment Tracking, Model Versioning, Automated Retraining, Probabilistic Forecasting, Energy Market Analysis, Distributed Systems, Cloud Infrastructure

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated deep learning models for metrological applications, improving model performance and training efficiency.
- Explored novel architectures and physics-informed hybrid loss functions to enhance model accuracy and stability.
- Collaborated with PhD researchers to publish findings at the FLOMEKO conference.
- Developed dashboards for data visualization and model performance monitoring.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Improved vitiligo segmentation accuracy to 94% and increased speed by 5x using advanced AI algorithms; deployed models with Docker and FastAPI.
- Developed a real-time tennis foul detection system using TensorRT, reducing compute cost and time by 6x.
- Managed MLOps pipelines on AWS SageMaker, addressing YOLO model data drift, and maintained Kubernetes infrastructure.
- Conducted educational sessions on machine learning and big data at universities, fostering knowledge sharing.
- Built POCs using YOLO, few-shot learning, and CNNs for client demonstrations.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Developed a full-stack website for AI-based services using Flask, HTML, and CSS.
- Built an AI chatbot application leveraging OpenAI API and large language models.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Built web scrapers and ETL pipelines for structured and unstructured data, enabling efficient data ingestion.
- Automated data cleaning workflows to streamline client analytics processes.

Projects

CAFA 6: Protein Function Prediction 2022

Deep Learning Competition

- Designed a weighted-probability ensemble integrating DPFunc and DeepGOPlus, leveraging AlphaFold for feature extraction.
- Enhanced model accuracy by 4% through custom feature engineering across MF, CC, and BP namespaces.

Razanlytics: Real-time Bitcoin Price Forecasting 2022

Thesis Project

- Developed a real-time forecasting model for Bitcoin prices using time series analysis and machine learning techniques.

Saaz: AI Self-Playing Chitralli Sitar 2023

Signal Processing, Music Generation

- Built the world's first AI-driven self-playing Chitralli Sitar, combining signal processing and music generation LLMs.
- Featured on Coke Studio in a music video by Hassan Raheem, showcasing innovative AI-driven music technology.

Hugging Face Open-Source

Contributor

- Co-authored the Zero-shot Image Classification technical documentation and implementation page on the Hugging Face Hub.

Education

Friedrich Alexander Universität Present

Master of Science in Artificial Intelligence Erlangen, DE

- Specialized in time series forecasting, MLOps, and distributed energy systems.

National University of Computer and Emerging Sciences Aug 2018 – June 2022

Bachelor of Science in Computer Science Karachi, PK

- Dean's List of Honor for academic excellence.
- Focused on machine learning, data engineering, and deep learning techniques.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python